

Checkpoint

12.5 Part 1: Lines

Problem 1. *What two things do you need for a line in 3D?*

Problem 2. *If I gave you a point on line L_1 and the equation of a parallel line L_2 , how would you find the equation of L_1 ?*

Problem 3. *Rewrite the equation of the following line in symmetric form.*

$$x(t) = 2t - 5, y(t) = -t + 7, z(t) = 3t - 4$$

Problem 4. *Rewrite the equation of the following line in parametric form.*

$$\frac{x+3}{-2} = \frac{y}{11} = z - 4$$

Problem 5. *Are the following lines intersecting, parallel, or skew?*

$$L_1 : x(t) = 2t + 5, y(t) = t - 3, z(t) = 2t$$

and

$$L_2 : \frac{x+1}{4} = \frac{y-4}{2} = \frac{z+2}{4}$$